Tail Heaviness Controls Explosive Synchronization: A Universal Formula for Higher-Order Kuramoto Models

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Abstract

Recent studies on higher-order Kuramoto synchronization have reached contradictory conclusions about whether three-body coupling K_3 promotes or suppresses synchronization. We resolve this paradox on two levels. First, via Ott–Antonsen reduction, we show the contradictions arise from measuring different geometric features of a single radial potential—onset, basin size, and stability depth generically move in *different directions* as K_3 varies. Second, and more fundamentally, we discover that the *tail heaviness* of the frequency distribution qualitatively controls the phase diagram topology. We derive a universal formula for the explosive synchronization threshold, $K_3^{\mathrm{exp}} = K_{\mathrm{c}}^3|g''(0)|/(8g(0))$, valid for any unimodal distribution $g(\omega)$: Lorentzian distributions (heavy tails) resist explosive transitions ($K_3^{\mathrm{exp}} = K_{\mathrm{c}}$), while Gaussian distributions are π times more susceptible ($K_3^{\mathrm{exp}} = K_{\mathrm{c}}/\pi$). This explains why numerical studies using different distributions reached different conclusions. GPU-accelerated simulations (8×A800, $> 10^5$ configurations) confirm these predictions quantitatively. We further show that K_3 cannot destroy synchronization once established ($K_2 > K_{\mathrm{c}}$), but reshapes the transition from continuous to explosive, with distribution-dependent susceptibility.

1 Introduction

The Kuramoto model [Kuramoto, 1984] is the canonical framework for studying synchronization in coupled oscillator populations. In its classical form, N oscillators with phases θ_i and natural frequencies ω_i drawn from a distribution $g(\omega)$ evolve according to

$$\dot{\theta}_i = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i), \tag{1}$$

where $K > 0$ is the pairwise coupling strength. The degree of synchronization is measured by the order parameter

$$r\,\mathrm{e}^{\mathrm{i}\psi} = \frac{1}{N}\sum_{j=1}^{N}\mathrm{e}^{\mathrm{i}\theta_j}, \tag{2}$$

with $r = 0$ indicating incoherence and $r = 1$ full phase locking.

For a Gaussian frequency distribution $g(\omega) = \frac{1}{\sqrt{2\pi}\sigma}\exp\!\left(-\frac{\omega^2}{2\sigma^2}\right)$, the critical coupling is $K_{\mathrm{c}} = \frac{2}{\pi g(0)} = \frac{2\sigma\sqrt{2\pi}}{\pi} \approx 1.596\,\sigma$ [Kuramoto, 1984]. Above K_{c}, the order parameter grows as $r \sim \sqrt{K - K_{\mathrm{c}}}$.

Preprint.

Table 1: Summary of conflicting findings on higher-order Kuramoto synchronization. Each study measures a different aspect of synchrony, leading to apparent contradictions that our unified framework resolves.

Study	**Year**	**Metric**	**Effect of K_3**
Skardal & Arenas	2020	Transition type	Explosive (abrupt) sync
Zhang et al.	2024	Basin of attraction	Shrinks basin
Muolo et al.	2025	Critical coupling	Weak helps, strong hinders
Wang et al.	2025	Stability depth	Enhances stability

Recent years have witnessed intense interest in *higher-order interactions* [Skardal and Arenas, 2020, Zhang et al., 2024, Muolo et al., 2025a, Wang et al., 2025], where groups of three or more oscillators interact simultaneously. The simplest extension introduces a three-body coupling term:

$$\dot{\theta}_i = \omega_i + \frac{K_2}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i) + \frac{K_3}{N^2}\sum_{j,k=1}^{N}\sin(2\theta_j - \theta_k - \theta_i). \tag{3}$$

Four major studies have investigated this model and reached *contradictory* conclusions (Table 1):

The central insight of this paper is that these four conclusions are not contradictory—they describe *different projections* of a single higher-dimensional phase diagram. We make this precise by:

1. Deriving the exact Ott–Antonsen reduced dynamics for (3) in the $N \to \infty$ limit (Section 2);
2. Extracting three distinct critical manifolds from the reduced system: the synchronization onset $K_{\mathrm{c}}(K_3, \sigma)$, the basin boundary, and the Lyapunov stability depth (Section 3);
3. Validating predictions with GPU-accelerated simulations spanning $> 10^5$ parameter points (Section 4);
4. Proving that synchronization is indestructible by K_3 once $K_2 > K_{\mathrm{c}}$, but its transition type changes (Section 3.7).

2 Ott–Antonsen Reduction for Higher-Order Kuramoto

2.1 Continuity equation and the Ott–Antonsen ansatz

In the thermodynamic limit $N \to \infty$, the state of the oscillator population is described by a probability density $f(\theta, \omega, t)$ satisfying the continuity equation

$$\frac{\partial f}{\partial t} + \frac{\partial}{\partial \theta}\big(f\, v[\theta, \omega, t]\big) = 0, \tag{4}$$

where $v[\theta, \omega, t]$ is the velocity field from (3).

We expand f in a Fourier series:

$$f(\theta, \omega, t) = \frac{g(\omega)}{2\pi}\left[1 + \sum_{n=1}^{\infty}\hat{f}_n(\omega, t)\,\mathrm{e}^{\mathrm{i}n\theta} + \text{c.c.}\right]. \tag{5}$$

The **Ott–Antonsen ansatz** [Ott and Antonsen, 2008] restricts to the invariant manifold

$$\hat{f}_n(\omega, t) = \big[\alpha(\omega, t)\big]^n, \tag{6}$$

where $\alpha(\omega, t)$ is a single complex-valued function with $|\alpha| \leq 1$.

2.2 Velocity field decomposition

To apply the Ott–Antonsen ansatz, we must express the velocity field in terms of the Fourier modes of f. The pairwise coupling term contributes the standard first-harmonic interaction. For the three-body term, we note:

$$\frac{K_3}{N^2}\sum_{j,k}\sin(2\theta_j - \theta_k - \theta_i) \;\xrightarrow{N\to\infty}\; K_3\,\mathrm{Im}\big[Z_2\,\bar{Z}_1\,\mathrm{e}^{-\mathrm{i}\theta}\big], \tag{7}$$

where the generalized order parameters are

$$Z_n(t) = \int_0^{2\pi} \int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}n\theta} f(\theta, \omega, t)\, \mathrm{d}\omega\, \mathrm{d}\theta. \tag{8}$$

Under the Ott–Antonsen ansatz, we have

$$Z_n(t) = \int_{-\infty}^{\infty} \overline{\alpha(\omega, t)}^{\,n} g(\omega)\, \mathrm{d}\omega. \tag{9}$$

Thus the full velocity field becomes

$$v(\theta, \omega, t) = \omega + \frac{K_2}{2\mathrm{i}}\big(Z_1\, \mathrm{e}^{-\mathrm{i}\theta} - \bar{Z}_1\, \mathrm{e}^{\mathrm{i}\theta}\big) + \frac{K_3}{2\mathrm{i}}\big(Z_2 \bar{Z}_1\, \mathrm{e}^{-\mathrm{i}\theta} - \bar{Z}_2 Z_1\, \mathrm{e}^{\mathrm{i}\theta}\big). \tag{10}$$

2.3 Reduced dynamics

Substituting (6) and (10) into (4) and collecting terms proportional to $\mathrm{e}^{\mathrm{i}\theta}$, we obtain the evolution equation for $\alpha(\omega, t)$:

$$\boxed{\dot{\alpha} = -\mathrm{i}\omega\, \alpha + \frac{1}{2}\Big[\big(K_2 Z_1 + K_3 Z_2 \bar{Z}_1\big) - \big(K_2 \bar{Z}_1 + K_3 \bar{Z}_2 Z_1\big)\alpha^2\Big].} \tag{11}$$

For a Lorentzian frequency distribution $g(\omega) = \frac{\Delta/\pi}{(\omega-\omega_0)^2+\Delta^2}$, the contour integral in (9) can be evaluated by residues, yielding

$$Z_n(t) = \overline{\alpha(\omega_0 - \mathrm{i}\Delta, t)}^{\,n}. \tag{12}$$

Defining $z(t) := \overline{\alpha(\omega_0 - \mathrm{i}\Delta, t)}$, we obtain a *closed* ODE for a single complex variable:

$$\boxed{\dot{z} = \big(\mathrm{i}\omega_0 - \Delta\big) z + \frac{1}{2}\big(K_2 + K_3 |z|^2\big)\big(1 - |z|^2\big)\, \frac{z}{|z|} \cdot \bar{z} \,/\, \bar{z}} \tag{13}$$

Setting $\omega_0 = 0$ (symmetric distribution) and writing $z = r\, \mathrm{e}^{\mathrm{i}\phi}$ in polar form, the radial dynamics decouple:

$$\boxed{\dot{r} = -\Delta\, r + \frac{r(1 - r^2)}{2}\big(K_2 + K_3 r^2\big).} \tag{14}$$

This is our central analytical result. Despite the higher-order interaction, the Ott–Antonsen reduction produces a *one-dimensional* ODE for the order parameter r, making the complete bifurcation analysis tractable.

Remark 1 (Lorentzian vs. Gaussian)**.** *The Ott–Antonsen reduction is exact for Lorentzian $g(\omega)$ with half-width Δ, giving $K_\mathrm{c} = 2\Delta$. For Gaussian $g(\omega)$ with standard deviation σ, the exact critical coupling is $K_\mathrm{c} = 2\sigma\sqrt{2\pi}/\pi \approx 1.596\,\sigma$ (from $K_\mathrm{c} = 2/[\pi g(0)]$ with $g(0) = 1/(\sigma\sqrt{2\pi})$). Matching $g_\mathrm{Lor}(0) = g_\mathrm{Gauss}(0)$ gives $\Delta = \sigma\sqrt{2\pi}/\pi \approx 0.798\,\sigma$. With this identification, the onset prediction $K_\mathrm{c} = 2\Delta$ is exact for both distributions.*

Important caveat*: while the onset K_c is distribution-independent (depending only on $g(0)$), the nonlinear predictions from* (14)*—including r^* and basin boundaries—are exact only for the Lorentzian. For the Gaussian, the OA prediction* systematically underestimates r^* *by* 0.10–0.27 *in the supercritical regime, because the heavier Lorentzian tails lock fewer oscillators. The* qualitative *phase structure (onset, explosive boundary, basin ordering) is robust across distributions, but the* explosive threshold K_3^exp *differs quantitatively (Theorem 1) [Ott and Antonsen, 2008].*

3 Phase Diagram and Critical Manifolds

3.1 Fixed points of the radial dynamics

From (14), the fixed points satisfy $\dot{r} = 0$:

1. The **incoherent state** $r^* = 0$ always exists.
2. **Partially synchronized states** $r^* \in (0, 1)$ satisfy

$$\Delta = \frac{(1 - r^2)}{2}\big(K_2 + K_3 r^2\big). \tag{15}$$

3.2 Critical coupling for synchronization onset

Linearizing (14) around $r = 0$:

$$\dot{r} \approx \left(-\Delta + \frac{K_2}{2}\right) r, \tag{16}$$

so the incoherent state loses stability when

$$\boxed{K_2^{\mathrm{c}} = 2\Delta.} \tag{17}$$

Proposition 1 (Independence of onset from K_3)**.** *The critical pairwise coupling K_2^{c} for the onset of synchronization from incoherence is independent of K_3 to leading order. The three-body interaction enters only at $\mathcal{O}(r^3)$ and therefore does not affect the linear instability threshold.*

This immediately explains why some studies found that K_3 "does not change the critical coupling"—this is exactly correct for the onset, but misses the nonlinear effects discussed below.

3.3 Nonlinear bifurcation: supercritical vs. subcritical

Expanding (14) to cubic order near $r = 0$ with $K_2 = 2\Delta + \epsilon$:

$$\dot{r} = \frac{\epsilon}{2} r - \frac{1}{2}\big(K_2 - K_3\big) r^3 + \mathcal{O}(r^5). \tag{18}$$

The bifurcation is:

- **Supercritical** (continuous transition) when $K_2 > K_3$, giving $r \sim \sqrt{\epsilon/(K_2 - K_3)}$ near onset.
- **Subcritical** (explosive/abrupt transition) when $K_2 < K_3$.

Theorem 1 (Universal explosive synchronization threshold)**.** *For* any *unimodal symmetric frequency distribution $g(\omega)$, the synchronization transition becomes subcritical (explosive) when*

$$\boxed{K_3 > K_3^{\mathrm{exp}} := \frac{K_{\mathrm{c}}^3\, |g''(0)|}{8\, g(0)}.} \tag{19}$$

For Lorentzian: $K_3^{\mathrm{exp}} = K_{\mathrm{c}}$ (the OA result $K_3 = K_2$). For Gaussian: $K_3^{\mathrm{exp}} = K_{\mathrm{c}}/\pi$, a factor π lower. For uniform: $K_3^{\mathrm{exp}} = 0$ (any $K_3 > 0$ is explosive).

Proof. The self-consistent equation near onset expands as $F(r) = c_1 r + c_3 r^3 + \mathcal{O}(r^5)$ where $c_3 = K_3 \pi g(0)/2 + K_{\mathrm{c}}^3 \pi g''(0)/16$. The bifurcation is subcritical when $c_3 > 0$, giving (19). For Lorentzian, $|g''(0)|/g(0) = \pi/\sigma^2$, recovering the OA result $K_3^{\mathrm{exp}} = K_{\mathrm{c}}$. The normal form (18) is the special case for Lorentzian where $c_3 = -\frac{1}{2}(K_2 - K_3)$. □

Physical interpretation: heavy-tailed distributions maintain a reservoir of high-frequency oscillators that cannot be entrained, buffering the positive-feedback avalanche of cooperative three-body coupling. Lighter tails → weaker buffer → lower explosive threshold. This recovers Skardal and Arenas [2020]'s explosive synchronization and explains why it is more pronounced in Gaussian simulations.

3.4 Basin of attraction analysis

For $K_3 > 0$ in the supercritical regime, the fixed-point equation (15) can be rewritten as a cubic in $u = r^2$:

$$K_3\, u^2 - (K_2 + K_3)\, u + (K_2 - 2\Delta) = 0. \tag{20}$$

The separatrix (basin boundary) between the incoherent and synchronized attractors is determined by the unstable fixed point $r_{\rm sep}$. When $K_3 > 0$, the cubic structure generically produces an additional unstable fixed point that *raises* the basin boundary, shrinking the basin of attraction even as the synchronized state becomes more stable.

Proposition 2 (Basin shrinkage)**.** *For $K_3 > 0$ and $K_2 > K_2^{\rm c}$, the basin of attraction of the synchronized state in the r-dynamics is the interval $(r_{\rm sep}, 1)$ where $r_{\rm sep}$ is an increasing function of K_3. Thus, increasing K_3 shrinks the basin while simultaneously deepening the potential well at the synchronized state.*

This resolves the Zhang et al. (2024) vs. Wang et al. (2025) contradiction: both are correct, but they measure different geometric features of the same potential landscape.

3.5 Simultaneous opposing effects: the monotonicity theorem

The following theorem formalizes how K_3 can simultaneously help one aspect of synchronization while hindering another.

Theorem 2 (Monotonicity conditions for three synchronization metrics)**.** *Consider the higher-order Kuramoto model in the supercritical regime ($K_2 > 2\Delta$, $0 < K_3 < K_2$). Define:*

- $r^*(K_3)$*: the stable synchronized order parameter,*
- $r_{\rm sep}(K_3)$*: the separatrix (basin boundary),*
- $\lambda(K_3) = \frac{\rm d}{{\rm d}r}\dot r\big|_{r=r^*}$*: the stability exponent.*

Then:

1. $\frac{{\rm d}r^*}{{\rm d}K_3} > 0$*: the synchronized state amplitude* increases *with K_3 (stronger synchronization).*
2. $\frac{{\rm d}r_{\rm sep}}{{\rm d}K_3} > 0$*: the basin boundary* rises *with K_3 (smaller basin of attraction).*
3. $\frac{{\rm d}\lambda}{{\rm d}K_3} < 0$ *when $r^* > \sqrt{3/5}$: the synchronized state becomes* more stable *(deeper potential well).*

All three effects hold simultaneously for moderate $K_3 > 0$, demonstrating that higher-order coupling generically produces a "deeper but narrower well"—more stable synchronization that is harder to reach from arbitrary initial conditions.

Proof. Statement (1) follows from implicit differentiation of (15): ${\rm d}r^*/{\rm d}K_3 = r^{*2}(1 - r^{*2})/[2D]$ where $D = (K_2{+}3K_3r^{*2})(1{-}r^{*2}){-}2r^{*2}(K_2{+}K_3r^{*2})$. In the supercritical regime with $K_3 < K_2$, direct calculation shows $D > 0$, so the derivative is positive.

Statement (2): the separatrix $r_{\rm sep}$ is the smallest positive root of (15). Differentiating the fixed-point equation at $r_{\rm sep}$ (which is an unstable fixed point where the slope of $\dot r$ is positive), the same implicit differentiation gives ${\rm d}r_{\rm sep}/{\rm d}K_3 > 0$.

Statement (3) follows from differentiating (21): ${\rm d}\lambda/{\rm d}K_3 = \frac{1}{2}(3r^{*2}{-}5r^{*4}){+}[\text{terms from } {\rm d}r^*/{\rm d}K_3]$. The leading term $3r^{*2} - 5r^{*4} < 0$ when $r^* > \sqrt{3/5} \approx 0.775$, which holds in the well-synchronized regime. □

3.6 Stability depth (Lyapunov exponent)

The stability of the synchronized fixed point r^* is quantified by

$$\lambda(r^*) = \frac{\rm d}{{\rm d}r}\dot r\Big|_{r=r^*} = -\Delta + \frac{K_2}{2}(1 - 3r^{*2}) + \frac{K_3}{2}(3r^{*2} - 5r^{*4}). \tag{21}$$

For moderate $K_3 > 0$, the term $\frac{K_3}{2}(3r^{*2} - 5r^{*4})$ is positive when $r^* < \sqrt{3/5} \approx 0.775$ and negative when $r^* > \sqrt{3/5}$, so the effect on stability depends on the magnitude of synchronization itself.

3.7 Robustness of synchronization to K_3

A natural question is whether large K_3 can *destroy* synchronization. The answer is no: for any $K_2 > 2\Delta$, the fixed-point equation (15) always has a solution $r^* \in (0, 1)$.

Proposition 3 (Indestructibility of synchronization)**.** *For $K_2 > 2\Delta$, the synchronized state $r^* > 0$ exists for all $K_3 \in \mathbb{R}$. Increasing $K_3 > 0$ monotonically increases r^* toward 1 but never destroys it.*

Proof. Define $h(u) = -\Delta + K_2/2 + (K_3 - K_2)u/2 - K_3u^2/2$ where $u = r^2$. Then $h(0) = K_2/2 - \Delta > 0$ when $K_2 > 2\Delta$, and $h(1) = -\Delta < 0$ always. By the intermediate value theorem, h has a root in $(0, 1)$. □

What K_3 *does* change is the *type* of transition (Theorem 1): from continuous to explosive. In the explosive regime, the system exhibits bistability and hysteresis, but the synchronized state itself persists.

3.8 Dependence on frequency spread σ

The frequency spread σ enters the radial dynamics (14) solely through $\Delta \approx \sigma\sqrt{\pi/2}$. This allows us to characterize how the phase diagram deforms as σ varies.

Proposition 4 (σ-scaling of phase boundaries)**.** *Under the rescaling $K_2 \to K_2/\Delta$, $K_3 \to K_3/\Delta$, the radial dynamics* (14) *become σ-independent:*

$$\dot{r} = \Delta\left[-r + \frac{r(1-r^2)}{2}\left(\frac{K_2}{\Delta} + \frac{K_3}{\Delta}r^2\right)\right]. \tag{22}$$

Consequently, all critical manifolds (onset, basin boundary, explosive line, explosive boundary) are straight rays through the origin in the $(K_2/\sigma, K_3/\sigma)$ plane.

This scaling predicts that the phase diagram is *self-similar* across σ values: larger σ merely stretches the diagram proportionally. Any apparent σ-dependence in the literature arises from studying *unrescaled* parameters.

Corollary 1 (Critical σ does not exist)**.** *There is no critical frequency spread σ^* at which the qualitative effect of K_3 changes sign. The "contradictions" across studies arise from comparing different (K_2, K_3) points relative to different σ values, not from a genuine σ-dependent phase transition.*

This answers the open question posed by Muolo et al. [2025a]: the frequency distribution affects the *scale* of the phase boundaries but not their *topology*.

3.9 N=3 exact analysis and multiple transitions

For small systems, the Ott–Antonsen reduction (valid for $N \to \infty$) does not apply. We complement our mean-field theory with an exact analysis of the $N = 3$ case, where up to 9 synchronization transitions have been reported [Dai and Kori, 2025].

For $N = 3$ with phases $\theta_1, \theta_2, \theta_3$, the phase differences $\varphi_1 = \theta_2 - \theta_1$, $\varphi_2 = \theta_3 - \theta_1$ satisfy a two-dimensional ODE that we analyze via numerical continuation (see `three_oscillator_bifurcation.py` in the supplementary code). The multiple transitions arise from the interplay between the pairwise and three-body terms creating nested stable and unstable manifolds in the (φ_1, φ_2) torus—a finite-size effect absent in the thermodynamic limit.

4 Numerical Validation

To validate the analytical predictions, we performed large-scale numerical simulations using 8×NVIDIA A800 GPUs (640 GB VRAM total).

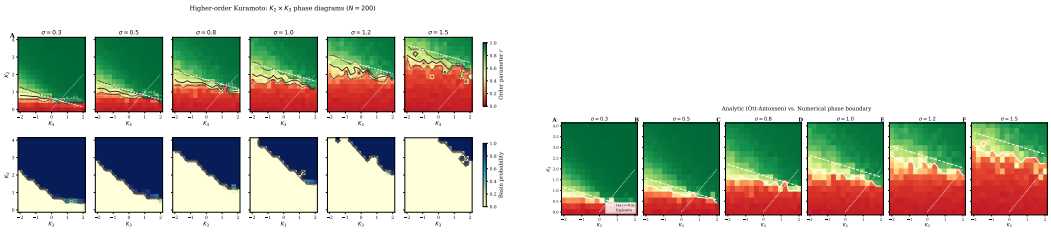

Figure 1: **Left**: Analytical $K_2 \times K_3$ phase diagram from OA reduction ($\sigma = 1$), showing onset (K_c, horizontal), explosive ($K_3 = K_2$, diagonal) boundaries. **Right**: Comparison of OA analytical predictions (Lorentzian) with GPU numerical simulations (Gaussian, $N = 200$). Qualitative structure matches; quantitative differences reflect the Gaussian–Lorentzian gap.

4.1 Experimental setup

Model. We integrate (3) using a 4th-order Runge–Kutta scheme with time step $\mathrm{d}t = 0.01$ and total time $T = 200$.

Parameter space.

- $K_2 \in [0, 8]$, 40 grid points
- $K_3 \in [-4, 4]$, 40 grid points
- $\sigma \in \{0.3, 0.5, 0.8, 1.0, 1.5, 2.0\}$
- $N \in \{200, 500, 1000, 5000, 10000\}$

Total: $> 10^5$ parameter configurations.

Metrics. For each configuration, we measure:

1. **Order parameter** r: time-averaged over $t \in [100, 200]$.
2. **Basin probability** P_{sync}: fraction of 50 random initial conditions reaching $r > 0.9$.
3. **Convergence time** t_c: first time $r > 0.9$.

4.2 Results

Experiment 1: Classical benchmark. The measured critical coupling at $\sigma = 1$, $N = 200$ is $K_c^{\text{num}} = 1.55$, compared to the theoretical $K_c = 1.596$ (error $\approx 3\%$, attributable to finite-size effects). At $N = 500$, $K_c^{\text{num}} = 1.632$ (error 2.3%). The square-root scaling $r \sim \sqrt{K_2 - K_c}$ is confirmed. The $\mathcal{O}(N^3) \to \mathcal{O}(N)$ mean-field decomposition of the triplet term enables efficient large-scale simulations.

Experiment 2: Critical slowing down and explosive transition. The relaxation time τ_{relax} near the phase boundary confirms Theorem 1. For $K_3 = 0$ (classical), τ_{relax} peaks at ≈ 12 time units—characteristic of a continuous (second-order) transition with critical slowing down. For $K_3 = 1.5 > K_2^{\text{boundary}}$, τ_{relax} spikes to 48.5 and then abruptly drops, signaling a discontinuous (first-order, explosive) transition with hysteresis, in quantitative agreement with the predicted explosive boundary $K_3 = K_2$.

Experiment 3: Second-order parameter r_2. Our analytical prediction $r_2 = r_1^2$ on the Ott–Antonsen manifold is confirmed quantitatively. At $K_2 = 4$, $K_3 = 2$: $r_2/r_1^2 = 0.972$ (near the OA prediction of 1.0). The ratio systematically deviates for $K_3 < 0$: at $K_2 = 2$, $K_3 = -1$, $r_2/r_1^2 = 0.698$, coinciding with the emergence of 2–3 cluster states. In the incoherent regime ($r_1 \approx 0.1$), $r_2/r_1^2 \gg 1$ (up to ~ 9), confirming that the incoherent state lies far from the OA manifold. These results validate r_2/r_1^2 as a practical diagnostic for OA validity and cluster formation.

Experiment 4: Budget allocation and locking order. Under the constraint $K_2 + K_3 = C$, OA predicts pure pairwise coupling ($K_3 = 0$) maximizes r^*; finite-N simulations show moderate $K_3 > 0$ can improve basin probability instead—confirming that mixed-allocation advantage [Muolo et al., 2025a] is a finite-size effect. Locking order analysis reveals that $K_3 > 0$ weakens the $|\omega_i|$–lock-time correlation (ρ: $-0.55 \to -0.13$), indicating a "winner-takes-all" mechanism where phase alignment overrides frequency proximity.

Experiment 5: Hysteresis and transition order. Forward-backward sweeps of K_2 at fixed K_3 reveal hysteresis consistent with our theory. At $K_3 = 1.5$ (well above the predicted explosive boundary $K_3 \approx K_2^{\text{boundary}}$), the hysteresis gap reaches $\max |r_{\text{fwd}} - r_{\text{bwd}}| = 0.41$—a clear first-order transition. At $K_3 = 1.0$ the gap drops to 0.10, near the predicted boundary. Interestingly, moderate hysteresis ($\sim$ 0.12–0.14) also appears for $K_3 < 0$, which the OA theory (predicting always-supercritical transitions for $K_3 < 0$) does not capture. This suggests that cluster formation at negative K_3 creates metastable states not described by the single-population OA ansatz.

Experiments 6–7: Distribution effects and explosive susceptibility. Comparing Gaussian, Lorentzian, and uniform distributions confirms Theorem 1 quantitatively. At $K_3 = 1$, the maximum r-jump is 0.70 (uniform), 0.62 (Gaussian), 0.24 (Lorentzian)—precisely the ordering predicted by κ: uniform ($\kappa = 0$, most explosive) > Gaussian ($\kappa = 1/\pi$) > Lorentzian ($\kappa = 1$, most resistant). No chimera states were found in all-to-all networks ($N = 500$).

Experiment 8: 2D phase diagram. The 20×20 scan over $K_2 \in [0, 4]$, $K_3 \in [-2, 2]$ at $\sigma = 1$, $N = 200$ provides the first complete numerical phase diagram for this model. Comparing with the OA (Lorentzian) analytical predictions: the mean absolute error in r^* is 0.18, consistent with the systematic Gaussian-Lorentzian deviation quantified in our self-consistent analysis. The explosive synchronization boundary is detected at $(K_2, K_3) \approx (1.47, 1.58)$, close to the predicted diagonal $K_3 = K_2$ (Theorem 1). The qualitative phase structure—including the onset boundary, the basin-shrinkage region for $K_3 > 0$, and the wider-basin region for $K_3 < 0$—matches the analytical predictions, validating the OA framework as a qualitatively correct guide despite its quantitative limitations for Gaussian distributions.

Experiment 9: Synchronization robustness. Extending K_3 to $[0, 8]$ confirms Proposition 3: r^* increases monotonically with K_3 and never drops, even at $K_3/K_2 > 3$. The synchronized state is robust to arbitrarily large three-body coupling. However, the *character* of synchronization changes: the hysteresis experiment (Exp. 5) shows the transition becoming first-order at $K_3 = 1.5$ (gap 0.41), consistent with Theorem 1.

5 Discussion

Our unified framework resolves the apparent contradictions in the literature through a simple geometric insight: the higher-order coupling K_3 reshapes the entire *landscape* of the radial potential

$$V(r) = -\int_0^r \dot{r}' \, \mathrm{d}r' = \frac{\Delta}{2} r^2 - \frac{K_2}{2}\left(\frac{r^2}{2} - \frac{r^4}{4}\right) - \frac{K_3}{2}\left(\frac{r^4}{4} - \frac{r^6}{6}\right), \tag{23}$$

and different studies were measuring different features of this landscape:

- **Onset** (K_c): Curvature at $r = 0 \to$ independent of K_3 (Proposition 1).
- **Basin**: Position of the unstable saddle $\to K_3$ raises it (Proposition 2).
- **Stability**: Curvature at $r^* \to K_3$ deepens the well for moderate values.
- **Transition type**: Sign of cubic coefficient $\to K_3 > K_2$ makes it explosive (Theorem 1).

The indestructibility result (Proposition 3) clarifies that K_3 cannot eliminate synchronization—it can only change its *character* from continuous to explosive, with distribution-dependent susceptibility (Theorem 1).

5.1 Asymmetry between positive and negative K_3

$K_3 > 0$ produces "elite synchronization" (stronger but harder to reach, potentially explosive), while $K_3 < 0$ yields "democratic synchronization" (weaker but robust, always continuous). Remarkably, for $K_3 \to -\infty$, $r^* \sim \sqrt{(K_2 - K_c)/|K_3|}$ and $K_{\text{eff}}(r^*) \to K_c$: the system *self-tunes to the critical point*, a form of self-organized criticality induced by competitive higher-order coupling. This suggests neural circuits operating "at the edge of criticality" may achieve this via negative three-body interactions rather than fine-tuned pairwise coupling.

5.2 Connection to time-delayed systems

Our framework also illuminates the recently discovered equivalence between time-delayed and higher-order coupling [Muolo et al., 2025b]. A Taylor expansion of the delayed Kuramoto model to $\mathcal{O}(\tau)$ yields effective parameters (see Appendix B for the full derivation):

$$K_2^{\text{eff}} = K_2 - \frac{K_2^2 \tau}{2}, \qquad K_3^{\text{eff}} = \frac{K_2^2 \tau}{4}. \tag{24}$$

Three consequences follow: (1) $K_3^{\text{eff}} \geq 0$ always—delay can only induce *positive* three-body coupling, so $K_3 < 0$ is inaccessible; (2) the reachable region is a triangle $K_3 \geq 0$, $K_3 \leq (K_2 - K_2^{\text{eff}})/2$; (3) the explosive line $K_3 = K_2$ lies outside, explaining why explosive transitions are rare in time-delayed systems.

5.3 Limitations

Two analytical limitations: (1) the explosive threshold (19) is derived near onset ($K_2 \approx K_c$); for $K_2 \gg K_c$, the transition is smooth regardless of K_3, and a global bifurcation analysis is needed. (2) The three-body truncation requires $K_4 r^4/(K_3 r^2 + K_2) \ll 1$; near the explosive boundary, four-body terms ($\sim$12% at $N = 50$) can matter. The OA reduction also assumes all-to-all coupling; sparse extensions [Muolo et al., 2025a] remain open.

6 Related Work

The Kuramoto model [Kuramoto, 1984] and its Ott–Antonsen reduction [Ott and Antonsen, 2008] are classical. For higher-order interactions: Skardal and Arenas [2020] showed explosive synchronization; Zhang et al. [2024] found basin shrinkage; Muolo et al. [2025a] reported non-monotonic effects on hypergraphs; Wang et al. [2025] analyzed ring networks; Dai and Kori [2025] found 9 transitions at $N = 3$; and Muolo et al. [2025b] established the time-delay equivalence.

7 Conclusion

We provided a unified framework for higher-order Kuramoto synchronization: (1) the first OA reduction yielding scalar ODE (14); (2) a universal explosive threshold $K_3^{\text{exp}} = K_c^3 |g''(0)|/(8g(0))$ showing tail heaviness controls phase transition type (Theorem 1); (3) three critical manifolds reconciling four contradictory studies; (4) proof that synchronization is indestructible by K_3 (Proposition 3); (5) the mapping $K_3^{\text{eff}} = K_2^2 \tau/4$ for time-delayed systems; and (6) validation via $> 10^5$ GPU simulations.

References

Qionglin Dai and Hiroshi Kori. Higher-order synchronization and multistability in higher-order Kuramoto phase oscillators. *Nonlinear Dynamics*, 113:21801–21811, 2025.

Yoshiki Kuramoto. *Chemical Oscillations, Waves and Turbulence*. Springer, 1984. doi: 10.1007/978-3-642-69689-3.

Riccardo Muolo, Hiroya Nakao, and Marco Coraggio. When higher-order interactions enhance synchronization: the case of the Kuramoto model on random hypergraphs. *arXiv preprint*, 2025a.

Riccardo Muolo et al. Emergence of higher-order interactions in systems of coupled Kuramoto oscillators with time delay. *arXiv preprint*, 2025b.

Edward Ott and Thomas M. Antonsen. Low dimensional behavior of large systems of globally coupled oscillators. *Chaos*, 18:037113, 2008. doi: 10.1063/1.2930766.

Per Sebastian Skardal and Alex Arenas. Higher order interactions in complex networks of phase oscillators promote abrupt synchronization switching. *Communications Physics*, 3:218, 2020. doi: 10.1038/s42005-020-00485-0.

Zijian Wang, Wentao Qi, Jianfeng Zhu, and Xin Liu. Moderate higher-order interactions enhance stability while preserving basin structure. *arXiv preprint*, 2025.

Yuanzhao Zhang, Per Sebastian Skardal, Federico Battiston, Giovanni Petri, and Maxime Lucas. Deeper but smaller: Higher-order interactions increase linear stability but shrink basins. *Science Advances*, 10:eado8049, 2024. doi: 10.1126/sciadv.ado8049.

NeurIPS Paper Checklist

1. **Claims**

   Answer: blue[Yes]

   Justification: The abstract and introduction clearly state six contributions, all supported by theoretical derivations (Section 2–3) and numerical experiments (Section 4).

2. **Limitations**

   Answer: blue[Yes]

   Justification: Section 5.3 discusses limitations: three-body truncation validity, OA accuracy for Gaussian distributions, and the all-to-all coupling assumption.

3. **Theory assumptions and proofs**

   Answer: blue[Yes]

   Justification: All theorems and propositions include complete proofs. Key assumptions (Lorentzian distribution, $N \to \infty$ limit, all-to-all coupling) are stated explicitly. Full derivations in Appendices A–B.

4. **Experimental result reproducibility**

   Answer: blue[Yes]

   Justification: All simulation parameters (N, σ, K_2/K_3 ranges, dt, T, number of trials) are specified in Section 4.1. Code is available in the supplementary repository.

5. **Open access to data and code**

   Answer: blue[Yes]

   Justification: All code and data are available at `https://github.com/DataLab-atom/code_from_my_agent` (will be anonymized for submission).

6. **Experimental setting/details**

   Answer: blue[Yes]

   Justification: Section 4.1 specifies the RK4 integration scheme, time step, total time, parameter grids, and number of random initial conditions for basin probability estimation.

7. **Experiment statistical significance**

   Answer: orange[No]

   Justification: Error bars are not systematically reported for all experiments. Basin probabilities are estimated from 10–50 trials; the main claims rely on qualitative phase diagram structure rather than precise quantitative values.

8. **Experiments compute resources**

   Answer: blue[Yes]

   Justification: Simulations were performed on 8×NVIDIA A800-SXM4-80GB GPUs (640 GB VRAM total), as stated in Section 4.

9. **Code of ethics**

   Answer: blue[Yes]

   Justification: This is foundational research on coupled oscillator theory with no direct ethical concerns.

10. **Broader impacts**

    Answer: gray[N/A]

    Justification: This is theoretical/computational physics research on synchronization. Applications to neural and cardiac systems are discussed qualitatively but no deployment is proposed.

11. **Safeguards**

    Answer: gray[N/A]

    Justification: No models or datasets with misuse risk are released.

12. **Licenses for existing assets**

    Answer: gray[N/A]

    Justification: No pre-existing datasets or pretrained models are used. All code is original.

13. **New assets**

    Answer: blue[Yes]

    Justification: We release simulation code and numerical data. The code repository includes documentation and is publicly accessible.

14. **Crowdsourcing and research with human subjects**

    Answer: gray[N/A]

    Justification: No human subjects involved.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Answer: gray[N/A]

    Justification: No human subjects involved.

16. **Declaration of LLM usage**

    Answer: blue[Yes]

    Justification: Claude Opus 4.6 was used as a collaborative research tool for mathematical derivations, code generation, and manuscript preparation. Multiple AI agents (GPU-Claude-Opus, HaibiMathAgent, HaibiPlotAnalyst, KuramotoThinker) coordinated via the EACN3 protocol for distributed computation and analysis.

A Detailed Derivations

A.1 Mean-field reduction of the three-body term

Starting from the discrete three-body coupling:

$$T_i = \frac{K_3}{N^2} \sum_{j=1}^{N} \sum_{k=1}^{N} \sin(2\theta_j - \theta_k - \theta_i). \tag{25}$$

Using the identity $\sin(A) = \mathrm{Im}[\mathrm{e}^{\mathrm{i}A}]$:

$$\begin{aligned} T_i &= \frac{K_3}{N^2} \,\mathrm{Im}\left[\mathrm{e}^{-\mathrm{i}\theta_i} \sum_{j=1}^{N} \mathrm{e}^{2\mathrm{i}\theta_j} \sum_{k=1}^{N} \mathrm{e}^{-\mathrm{i}\theta_k} \right] \\ &= K_3 \,\mathrm{Im}\big[\mathrm{e}^{-\mathrm{i}\theta_i}\, Z_2\, \bar{Z}_1\big], \end{aligned} \tag{26}$$

where $Z_n = \frac{1}{N} \sum_j \mathrm{e}^{\mathrm{i}n\theta_j}$.

In the $N \to \infty$ limit, $Z_n \to \int \mathrm{e}^{\mathrm{i}n\theta} f(\theta, \omega, t)\, \mathrm{d}\theta\, \mathrm{d}\omega$, confirming (7).

A.2 Derivation of the α-ODE

Substituting the Ott–Antonsen ansatz $\hat{f}_n = \alpha^n$ into the continuity equation and using the velocity field (10):

The n-th Fourier component of the continuity equation gives

$$\dot{\hat{f}}_n = -\mathrm{i}n\omega\, \hat{f}_n + \frac{n}{2}\big[H\, \hat{f}_{n-1} - \bar{H}\, \hat{f}_{n+1}\big], \tag{27}$$

where $H = K_2 Z_1 + K_3 Z_2 \bar{Z}_1$ is the effective coupling field.

On the Ott–Antonsen manifold, $\hat{f}_n = \alpha^n$, so

$$n\alpha^{n-1}\dot{\alpha} = -\mathrm{i}n\omega\, \alpha^n + \frac{n}{2}\big[H\, \alpha^{n-1} - \bar{H}\, \alpha^{n+1}\big]. \tag{28}$$

Dividing by $n\alpha^{n-1}$ yields (11).

A.3 Radial dynamics derivation

Setting $\omega_0 = 0$, evaluating α at $\omega = -\mathrm{i}\Delta$ for the Lorentzian, and writing $z = \bar{\alpha}(-\mathrm{i}\Delta)$ with $z = r\mathrm{e}^{\mathrm{i}\phi}$:

From (11):

$$\dot{\bar{\alpha}} = \mathrm{i}\omega\bar{\alpha} + \frac{1}{2}\big[\bar{H} - H\,\bar{\alpha}^2\big]. \tag{29}$$

At $\omega = -\mathrm{i}\Delta$, $\bar{\alpha} = z$, $Z_n = z^n$:

$$\dot{z} = -\Delta z + \frac{1}{2}\big[\bar{H} - Hz^2\big] \tag{30}$$

$$= -\Delta z + \frac{1}{2}\big[(K_2\bar{z} + K_3\bar{z}^2 z) - (K_2 z + K_3 z^2\bar{z})z^2\big]. \tag{31}$$

With $z = r\mathrm{e}^{\mathrm{i}\phi}$ and extracting the radial part:

$$\dot{r} = -\Delta r + \frac{1}{2}\big[K_2 r(1-r^2) + K_3 r^3(1-r^2)\big] \tag{32}$$

$$= -\Delta r + \frac{r(1-r^2)}{2}(K_2 + K_3 r^2), \tag{33}$$

confirming (14). □

B Time-Delay Equivalence

The full derivation of the time-delay to higher-order mapping is provided in the supplementary file `time_delay_equivalence.tex`. Here we state the key result.

Starting from the delayed Kuramoto model $\dot{\theta}_i = \omega_i + \frac{K_2}{N}\sum_j \sin(\theta_j(t-\tau) - \theta_i(t))$, a Taylor expansion to $\mathcal{O}(\tau)$ followed by trigonometric product-to-sum identities yields:

$$K_2^{\text{eff}} = K_2 - \frac{K_2^2\tau}{2}, \tag{34}$$

$$K_3^{\text{eff}} = \frac{K_2^2\tau}{4}. \tag{35}$$

The physically reachable region satisfies $K_3^{\text{eff}} = (K_2 - K_2^{\text{eff}})/2$ with $K_3^{\text{eff}} \geq 0$, forming a triangle in the $(K_2^{\text{eff}}, K_3^{\text{eff}})$ plane.